



SWAMINARAYAN UNIVERSITY

Program Name: Engineering

Level: Diploma

Course/ Subject Code: DE100101

Course/Subject Name: Basic Mathematics

Question 1

If a matrix A has an order of $m \times n$ and it is defined as a column matrix, which of the following mathematical conditions must be true?

- (A) $m = 1$
- (B) $n = 1$
- (C) $m = n$
- (D) $m \neq n$

Question 2

Given two matrices $A = \begin{bmatrix} 3 & -1 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -3 & 2 \end{bmatrix}$, what is the resulting matrix for the operation $2A - B$?

- (A) $\begin{bmatrix} 5 & -6 \\ 7 & 8 \end{bmatrix}$
- (B) $\begin{bmatrix} 7 & -6 \\ 1 & 12 \end{bmatrix}$
- (C) $\begin{bmatrix} 5 & -6 \\ 1 & 8 \end{bmatrix}$
- (D) $\begin{bmatrix} 5 & 2 \\ 7 & 8 \end{bmatrix}$

Question 3 (Idea of Determinants)

Which of the following descriptions accurately characterizes a determinant?

- (A) It is a rectangular array of numbers that can be simplified into a single algebraic value.
- (B) It is a unique scalar number associated exclusively with a square matrix.
- (C) It is a matrix operation that can be executed on matrices of any arbitrary order $m \times n$.
- (D) It is an identical copy of a diagonal matrix scaled by a specific constant factor.

Question 4

For the matrix $M = \begin{bmatrix} 4 & -2 \\ 3 & 7 \end{bmatrix}$, what are the exact values of the minor M_{12} and the cofactor C_{12} for the element in the first row and second column?

- (A) $M_{12} = 3$ and $C_{12} = 3$
- (B) $M_{12} = -2$ and $C_{12} = 2$
- (C) $M_{12} = 3$ and $C_{12} = -3$
- (D) $M_{12} = 7$ and $C_{12} = -7$

Question 5

What is the unique solution vector $\begin{bmatrix} x \\ y \end{bmatrix}$ for the system of linear equations represented by the matrix equation $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$?

- (A) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$
- (B) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- (C) $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$
- (D) $\begin{bmatrix} 0 \\ 5 \end{bmatrix}$

Answer

1. B
2. A
3. B
4. C
5. A

Question 1

Find the area of a trapezium whose parallel sides are 12 cm and 18 cm, and the perpendicular distance between them is 10 cm.

- A) 120 square cm
- B) 150 square cm
- C) 300 square cm
- D) 180 square cm

Question 2

A right circular cylinder has a base radius of 7 cm and a height of 10 cm. What is its total surface area? (Take $\pi = 22/7$)

- A) 748 square cm
- B) 440 square cm
- C) 616 square cm
- D) 880 square cm

Question 3

Calculate the volume of a cone with a base radius of 6 cm and a vertical height of 14 cm. (Take $\pi = 22/7$)

- A) 264 cubic cm
- B) 528 cubic cm
- C) 1584 cubic cm
- D) 792 cubic cm

Question 4

What is the surface area of a solid sphere whose radius is 7 cm? (Take $\pi = 22/7$)

- A) 154 square cm
- B) 308 square cm
- C) 616 square cm
- D) 1232 square cm

Question 5

A rectangular cuboid has a length of 5 cm, a width of 4 cm, and a height of 3 cm. Find its total surface area.

- A) 47 square cm
- B) 60 square cm
- C) 94 square cm
- D) 120 square cm

Answer

- 1. B**
- 2. A**
- 3. B**
- 4. C**
- 5. C**

1.

What is the radian measure of an angle of -210° ?

- (a) $-\frac{7\pi}{6}$
- (b) $-\frac{5\pi}{6}$
- (c) $-\frac{4\pi}{3}$
- (d) $-\frac{5\pi}{4}$

2.

If $\cos(\alpha + \beta) = 0$, then the expression $\sin(\alpha - \beta)$ can be simplified to which of the following?

- (a) $\sin 2\alpha$
- (b) $\cos \beta$
- (c) $\cos 2\beta$
- (d) $\sin \beta$

3.

Which of the following expressions is identically equal to $\frac{1-\cos 2x}{\sin 2x}$ for all values of x where the expression is defined?

- (a) $\cot x$
- (b) $\tan x$
- (c) $\sin x$
- (d) $\cos x$

4.

What is the fundamental period of the trigonometric function $f(x) = 3\cos\left(\frac{4x}{5}\right) + 2$?

- (a) 2π
- (b) $\frac{5\pi}{2}$
- (c) $\frac{5\pi}{4}$
- (d) 5π

5. Inverse Trigonometric Functions

What is the principal value of $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$?

- (a) $\frac{2\pi}{3}$
- (b) $\frac{\pi}{3}$
- (c) $-\frac{\pi}{3}$
- (d) $\frac{4\pi}{3}$

Answer 1) A 2) C 3) B 4) B 5) B

Question 1:

What is the unit vector $\hat{u}$ in the direction of the vector $\vec{A} = 3\hat{i} - 4\hat{j}$?

- A) $\frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$
- B) $3\hat{i} - 4\hat{j}$
- C) $\frac{3}{25}\hat{i} - \frac{4}{25}\hat{j}$
- D) $5\hat{i} - 5\hat{j}$

Question 2:

If two vectors $\vec{P} = 2\hat{i} + 3\hat{j}$ and $\vec{Q} = 4\hat{i} - 5\hat{j}$ are added together, what is the modulus (magnitude) of their resultant vector $\vec{R} = \vec{P} + \vec{Q}$?

- A) $\sqrt{10}$
- B) $\sqrt{40}$
- C) $2\sqrt{10}$
- D) 8

Question 3:

What is the angle θ between the vectors $\vec{A} = \hat{i} + \hat{j}$ and $\vec{B} = \hat{i} - \hat{j}$?

- A) 0°
- B) 45°
- C) 90°
- D) 180°

Question 4:

Which of the following physical quantities is calculated using the cross product (vector product) of two vectors?

- A) Work done by a force
- B) Power
- C) Electric flux
- D) Torque

Question 5:

A constant force $\vec{F} = (2\hat{i} + 3\hat{j} - \hat{k})$ N displaces an object from the origin to a position $\vec{d} = (4\hat{i} + 2\hat{j} + 5\hat{k})$ m. What is the total work done by the force?

- A) 19 J
- B) 9 J
- C) 11 J
- D) 15 J

Answer

1. A
2. B
3. C
4. D
5. B

Q1.

If $3^{2x-1} = 27$, which of the following represents the correct logarithmic form and the correct value of x ?

A) $\log_3(27) = 2x - 1$, and $x = 2$

B) $\log_{27}(3) = 2x - 1$, and $x = 1$

C) $\log_3(2x - 1) = 27$, and $x = 2$

D) $\log_{2x-1}(27) = 3$, and $x = 1$

Q2.

Which of the following statements is **incorrect** regarding the working rules of logarithms for valid bases and arguments ($a > 0$, $a \neq 1$, and $M, N > 0$)?

A) $\log_a(M \cdot N) = \log_a M + \log_a N$

B) $\frac{\log_a M}{\log_a N} = \log_a(M - N)$

C) $\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$

D) $\log_a(M^p) = p \cdot \log_a M$

ANSWER 1. A 2.B